

Weekly Activity Log

Roll No:

Student Name:

Project Guide Name:

Week	Date	Activity Performed	Outcome	Student Sign	Guide Sign
1		Project discussion and requirement gathering- Project title with 4 objectives	Project requirements finalized		
2		Study of existing system and problem identification	Existing system analyzed		
3		Software requirement analysis	Functional requirements prepared		
4		Database design (ER Diagram)	Database schema completed		
5		UI design (Wireframes/Mockups)	Screen layouts prepared		
6		Module 1 development	First module completed		
7		Module 2 development	Second module completed		
8		Database integration	Database connected with application		
9		Testing of modules	Bugs identified and fixed		
10		User Interface improvements	Application refined		
11		Final testing and documentation	Project documentation completed		
12		Project demonstration and submission	Final project submitted		

ORDER FOR PROJECT REPORT:

Title Page ☒

Certificate ☒

Certificate by the Organization

Declaration ☒

Acknowledgement

Abstract

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The student would be required to adhere to the following standards for the submission of project report:

1. 1 Sides printing
2. 8. Vertical space between paragraphs shall be about 1.5 line spacing.

Preferred

- Header Font size: 14 pts
- Normal content Font size: 12 pts
- Font Face: Times New Roman or Arial font

3. Margins

- Left: 1.5 inches
- Right : 1.0 inches
- Top : 1.0 inches
- Bottom : 1.0 inches

4. Page size A4

5. Color of report should be black

6. Reference only in APA Style

7. Page Number shall begin from Chapter1 till Chapter 8.

AutoCare Hub

A Project Report

Submitted in partial fulfillment of the requirements for the award
of the degree of

BACHELOR OF SCIENCE

COMPUTER SCIENCE

By

Norafsha Shaikh
Roll Number : 563

Under the esteemed guidance of

Khushi Mam

Assistant Professor



NAGINDAS KHANDWALA COLLEGE

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MAHARASHTRA

2026 – 27

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DEPARTMENT OF COMPUTER AND INFORMATION SCIENCE



CERTIFICATE

This is to certify that the project titled, **AutoCare Hub** is bonafied work of **Norafsha Shaikh** bearing Roll No: 563 is submitted in partial fulfillment of the requirements for the award of degree of BACHELOR OF SCIENCE in COMPUTER SCIENCE from University of Mumbai.

Internal Guide

Coordinator

External Examiner

Date

College Seal

DECLARATION

I hereby declare that the project entitled, **PAPER FACTORY ERP MANAGEMENT** done at **Nagindas Khandwala College**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university. The project is done in partial fulfillment of the requirements for the award of degree of **BACHELOR OF SCIENCE** to be submitted as final semester project as part of our curriculum.

Signature

Norafsha Shaikh

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without expressing gratitude to the people who made it possible. I take immense pleasure in presenting this project report on "**AutoCare Hub**".

First and foremost, I express my deepest gratitude to my esteemed project guide, **Prof. Khushi Mam** Assistant Professor, Department of Computer and Information Science, Nagindas Khandwala College, for her invaluable guidance, constructive suggestions, constant encouragement, and insightful feedback throughout the duration of this project.

I would also like to extend my heartfelt thanks to the **Head of the Department**, the respected **Principal**, and the management of **Nagindas Khandwala College** for providing the requisite infrastructure, laboratory resources, and academic environment conducive to rigorous project work.

Finally, I would like to thank my family, peers, and industry professionals whose continuous moral support, technical discussions, and cooperation were instrumental in bringing this project to fruition.

Norafsha Shaikh

Roll No: 563

B.Sc. Computer Science

CHAPTER 1: INTRODUCTION

1.1 Project Overview

AutoCare Hub is a web-based Garage Management System designed to simplify and automate garage operations. It helps administrators manage **customers, vehicles, service jobs, and service records** in one centralized platform.

The system is developed using **React.js with Vite** for the frontend, **Node.js and Express.js** for the backend, and **MySQL** for database management. It includes secure administrator login and a dashboard for viewing important garage information.

The main objective of AutoCare Hub is to reduce manual paperwork, improve record management, save time, and provide an efficient and user-friendly solution for managing garage activities.

1.2 Problem Statement

Traditional garage management often relies on manual records, which can lead to **data loss, errors, time consumption, and difficulty in tracking service details**. Managing customer information, vehicle records, and service jobs manually can also make it difficult to monitor pending services and maintain accurate records.

Therefore, there is a need for a **digital and centralized garage management system** that can efficiently manage customer, vehicle, and service information while reducing manual work. **AutoCare Hub** is developed to address these problems by providing a simple, organized, and user-friendly platform for managing garage operations.

1.3 Objectives

The main objective of **AutoCare Hub** is to develop a user-friendly and efficient web-based system for managing the daily operations of an automobile garage. The system aims to digitally maintain and organize customer information, vehicle details, and service job records in a centralized database. It helps administrators easily add, update, search, and manage records while reducing manual paperwork.

and the chances of errors or data loss. Another objective is to provide a dashboard for monitoring important garage information, including customers, vehicles, completed services, and pending service jobs. The system also aims to improve the overall efficiency of garage operations, save time, provide quick access to records, and make the management process more organized, accurate, and convenient.

1.4 Scope of the Project

The scope of **AutoCare Hub** covers the digital management of major activities involved in an automobile garage. The system allows administrators to manage customer details, vehicle information, and service job records through a centralized platform. It provides features for adding, updating, searching, and viewing records, along with tracking pending and completed services. The system also includes an administrator login and dashboard to provide quick access to important garage information. By reducing manual record-keeping, AutoCare Hub helps improve accuracy, efficiency, and organization. The project can also be extended in the future by adding features such as online service booking, billing and payment management, service reminders, notifications, inventory management, and customer accounts.

CHAPTER 2: ORGANIZATION PROFILE AND FIELD STUDY

2.1 About the Organization

ORGANIZATION NAME: Auto Motors

ADDRESS: Mumbai, Maharashtra, India

TYPE OF ORGANIZATION: Automobile Service and Repair Garage

NATURE OF ORGANIZATION: Automobile Servicing, Repair and Maintenance

YEAR OF ESTABLISHMENT: 2015

2.2 Organization Structure

The organization is divided into different functional areas, with each department responsible for specific activities. The major departments considered in the proposed system are:

- **Management:** Oversees the overall functioning of the garage and makes important operational decisions.
- **Service Department:** Handles vehicle servicing, maintenance, repairs, and service-related activities.
- **Customer Management:** Maintains customer information and manages customer requirements and service requests.
- **Vehicle Management:** Maintains vehicle details, service history, and other vehicle-related records.
- **Accounts Department:** Handles billing, payment, and other financial records.
- **Administration:** Supports the overall coordination, record maintenance, and management of garage activities.

The AutoCare Hub system will connect these major activities through a centralized platform, making it easier to maintain and access customer, vehicle, and service information.

2.3 Services / Products Offered

AutoCare Hub Garage primarily focuses on **vehicle servicing, repair, and maintenance activities**. The organization provides various services according to the requirements of vehicle owners.

The major activities include:

- Vehicle servicing and maintenance
- Vehicle inspection and diagnosis
- Repair and replacement of vehicle components
- Maintaining customer and vehicle records
- Managing service jobs and service status
- Tracking pending and completed services
- Maintaining service history
- Handling customer service requirements

The AutoCare Hub system will help maintain records related to these activities and provide better visibility into customer vehicles and ongoing service operations

2.4 Existing Business Process

AutoCare Hub Garage primarily focuses on **vehicle servicing, repair, and maintenance activities**. The organization provides various services according to the requirements of vehicle owners.

The major activities include:

- Vehicle servicing and maintenance
- Vehicle inspection and diagnosis

- Repair and replacement of vehicle components
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- Managing service jobs and service status
- Tracking pending and completed services
- Maintaining service history
- Handling customer service requirements

The AutoCare Hub system will help maintain records related to these activities and provide better visibility into customer vehicles and ongoing service operations

2.5 Problem Identified During Field Study

Field observations identified several difficulties in the traditional garage management process. Customer and vehicle details were often maintained using **manual records**, making information difficult to search and update. Service jobs and pending services were also difficult to track, which could result in delays and missed service activities.

Maintaining service history manually may lead to **data duplication, incorrect entries, and loss of important records**. Employees may also spend considerable time searching through registers and updating records. The absence of a centralized system makes it difficult for management to monitor the overall status of customers, vehicles, and service jobs efficiently.

Therefore, a centralized digital system such as **AutoCare Hub** is required to improve record management, reduce manual work, track service activities efficiently, and provide quick access to important garage information.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Existing System

The existing garage management system mainly relies on **manual record-keeping**, such as registers, spreadsheets, and separate files. Customer details, vehicle information, and service records are maintained manually, making it difficult to search, update, and track information. Tracking pending and completed service jobs can also be time-consuming and may lead to errors or loss of records. This creates a need for a centralized digital system like **AutoCare Hub** to manage garage operations more efficiently.

3.2 Proposed System

The proposed **AutoCare Hub** is a web-based Garage Management System developed to digitize and simplify the daily operations of an automobile garage. The system provides a centralized platform for managing **customer details, vehicle information, service jobs, service history, and pending services**. Administrators can easily add, update, search, and view records through a user-friendly interface. A dashboard provides an overview of important garage activities and helps in monitoring service operations. The system is developed using **React.js with Vite, Node.js, Express.js, and MySQL**. By replacing manual record-keeping with a digital system, AutoCare Hub helps reduce paperwork, minimize errors, save time, and improve the overall efficiency and organization of garage management.

3.3 Functional Requirements

The functional requirements of **AutoCare Hub** describe the main features and operations that the system must perform. The system should provide the following functionalities:

1. **Admin Login:** The system should allow authorized administrators to log in using their email and password.
2. **Dashboard Management:** The system should display important information such as total customers, vehicles, service jobs, and pending services.
3. **Customer Management:** The administrator should be able to add, view, update, search, and manage customer details.
4. **Vehicle Management:** The system should allow the administrator to store and manage vehicle information such as vehicle number, model, and customer details.
5. **Service Job Management:** The administrator should be able to create, update, view, and manage service jobs for vehicles.
6. **Service Status Tracking:** The system should allow service jobs to be tracked based on their status, such as pending, in progress, or completed.
7. **Search and Record Management:** The system should provide options to search and access customer, vehicle, and service records quickly.
8. **Profile Management:** The administrator should be able to view and update profile information such as name, email, and password.
9. **Database Management:** The system should securely store and retrieve customer, vehicle, and service information from the MySQL database.
10. **Logout:** The system should provide a logout function to securely end the administrator's session.

3.4 Non-Functional Requirements

- **Performance:** The system should provide quick response times when adding, updating, searching, and retrieving customer, vehicle, and service records.
- **Security:** The system should ensure that only authorized administrators can access the system and its management features.

- **Usability:** The application should have a simple, clear, and user-friendly interface that allows administrators to perform tasks easily.
- **Reliability:** The system should function consistently and maintain accurate customer, vehicle, and service information.
- **Scalability:** The system should be capable of handling an increasing number of customers, vehicles, and service records.
- **Maintainability:** The system should be easy to maintain, modify, and upgrade when new features or requirements are introduced.
- **Responsiveness:** The application should work properly on different screen sizes and devices, providing a consistent user experience.
- **Data Integrity:** The system should maintain accurate and consistent data while storing and retrieving records from the MySQL database.

3.5 System Architecture

The **AutoCare Hub** follows a **three-tier architecture** consisting of the **Presentation Layer, Application Layer, and Database Layer**. The Presentation Layer is developed using **React.js with Vite** and provides the user interface through which the administrator interacts with the system. The Application Layer uses **Node.js and Express.js** to handle business logic, authentication, requests, and communication between the frontend and database. The Database Layer uses **MySQL** to store and manage customer, vehicle, service, and administrator information.

The overall architecture can be represented as:

Administrator → React.js (Frontend) → Node.js + Express.js (Backend/API) → MySQL Database

This architecture provides a structured and centralized system, making AutoCare Hub easier to manage, maintain, and extend in the future.

3.6 Database Design

The **AutoCare Hub** database is designed using **MySQL** to store and manage all important information related to garage operations. The database follows a structured approach to maintain accurate and organized records and to establish relationships between customers, vehicles, and service jobs.

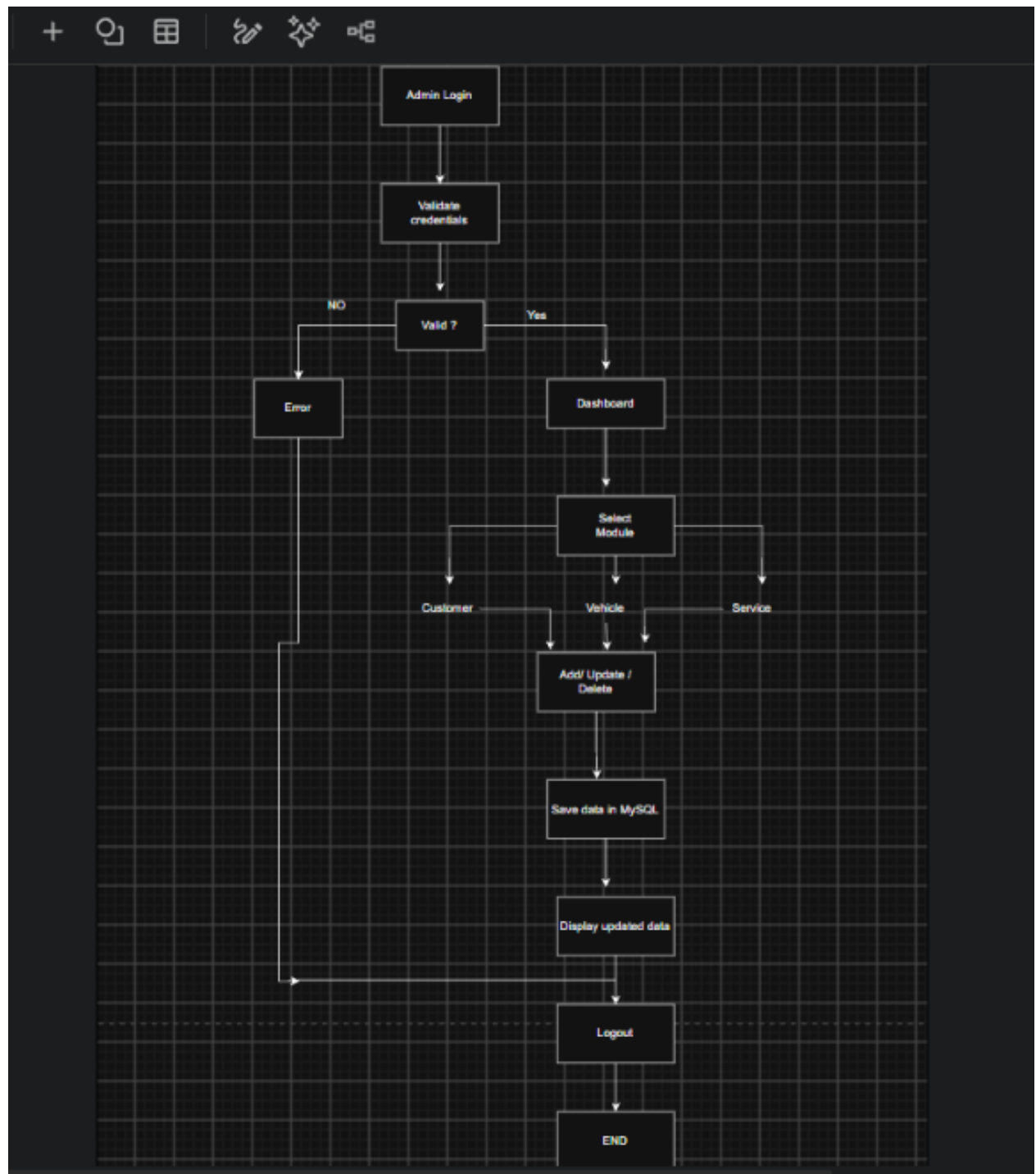
The major tables used in the system are:

- **Admin:** Stores administrator details such as admin ID, name, email, and password for login and profile management.
- **Customers:** Stores customer information such as customer ID, name, phone number, and address.
- **Vehicles:** Stores vehicle details such as vehicle ID, customer ID, vehicle number, and vehicle model.
- **Services:** Stores service job information such as service ID, vehicle ID, service details, service date, and service status.

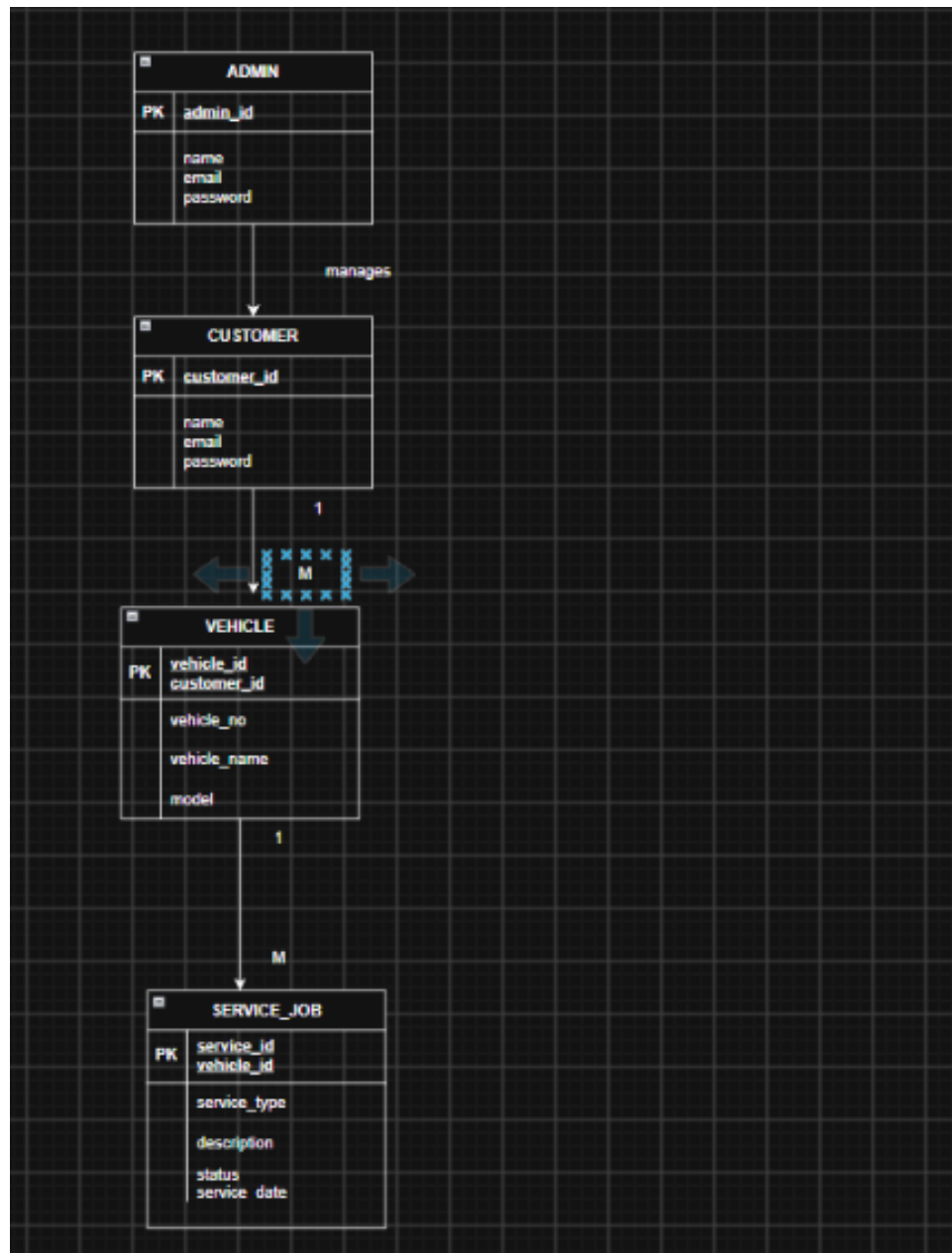
The relationship between the tables allows a customer to have one or more vehicles, while each vehicle can have multiple service records. This database design helps maintain **data consistency, easy data retrieval, and centralized record management** for the AutoCare Hub system.

3.7 UML Diagrams

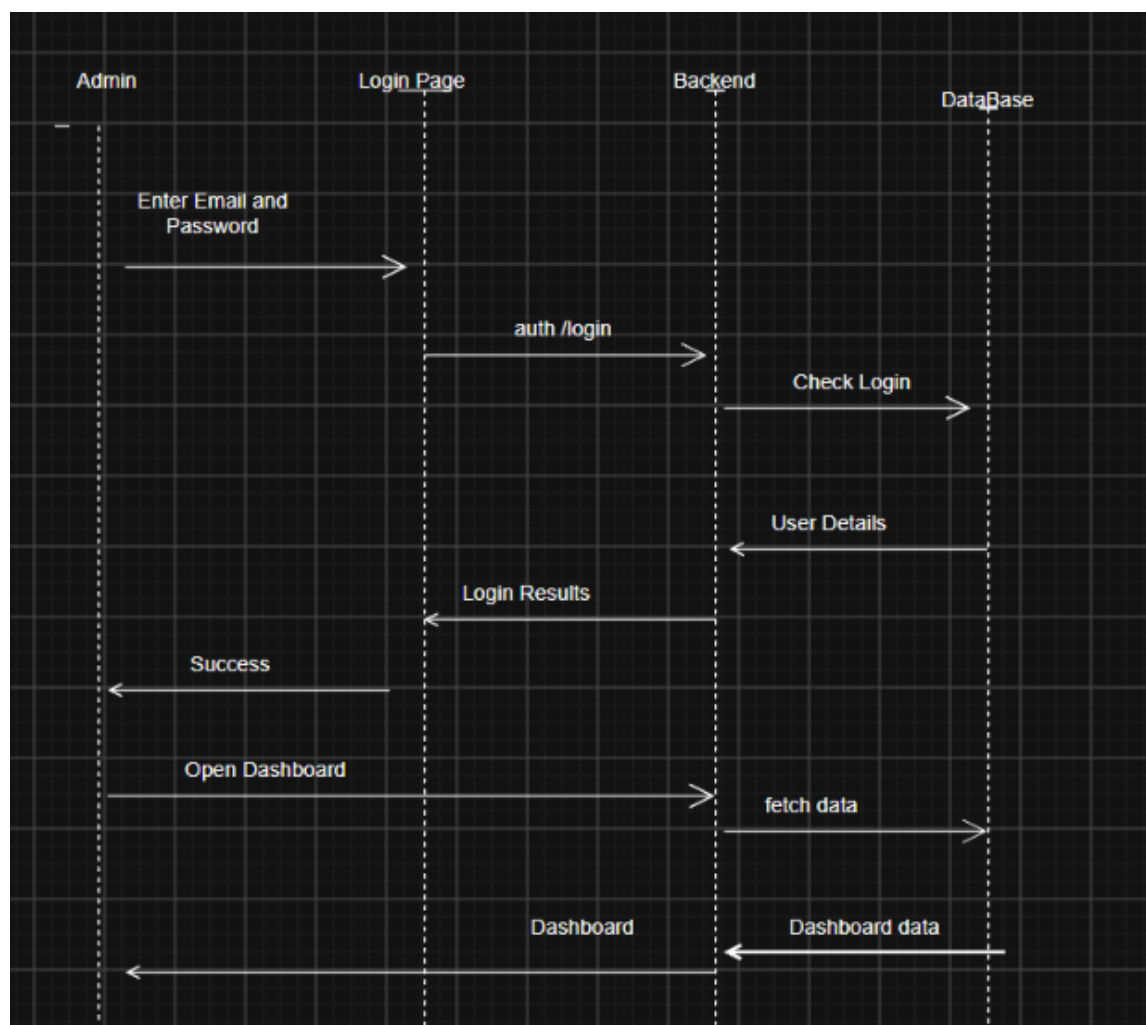
Activity diagram:



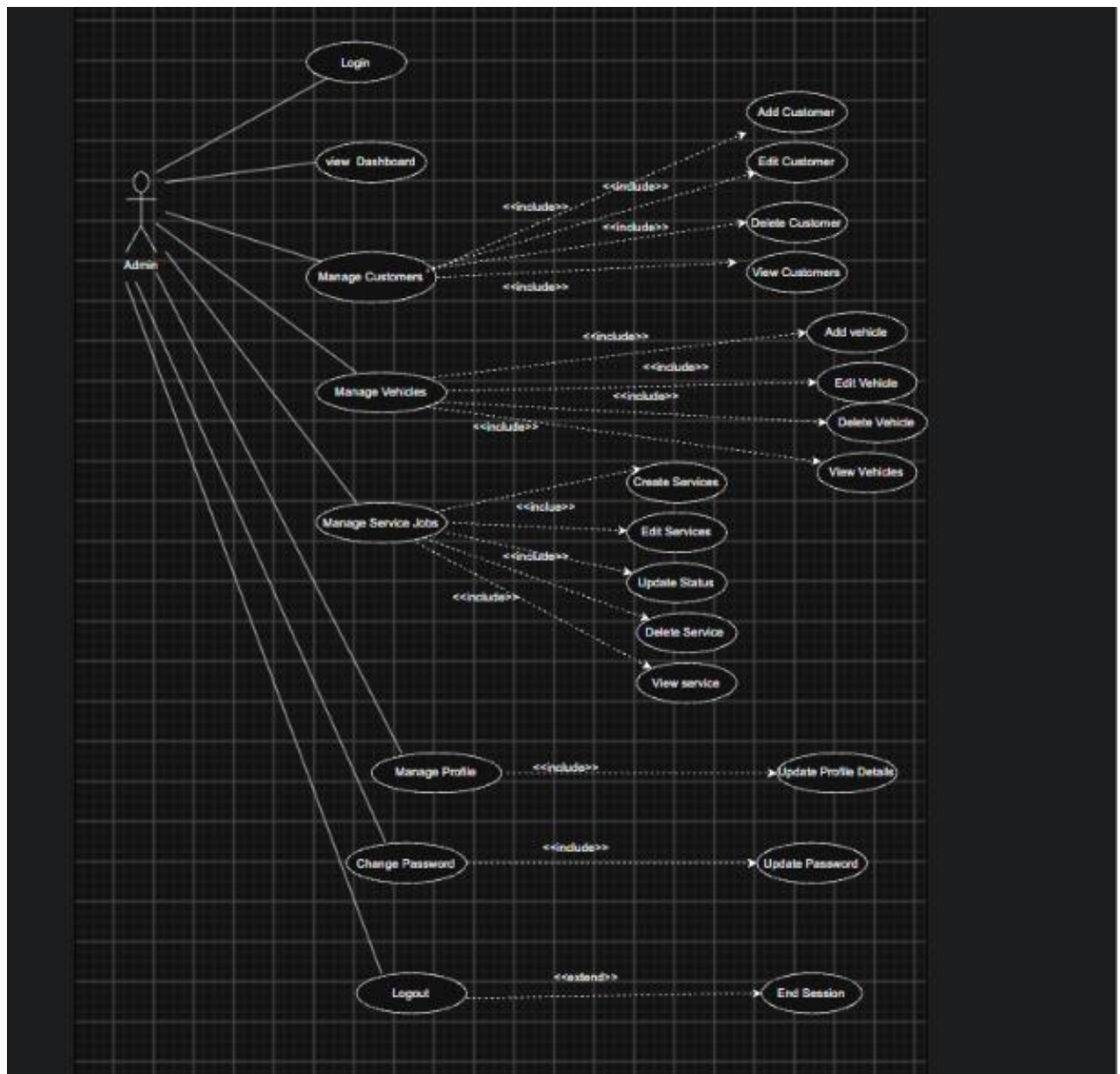
ER Diagram:



Sequence diagram :



USE CASE DIAGRAM:



CLASS DIAGRAM:



CHAPTER 4: SYSTEM IMPLEMENTATION

4.1 Hardware Setup

The **AutoCare Hub** system requires basic computer hardware to run the frontend, backend, and MySQL database smoothly. The hardware setup used for the project includes:

Hardware Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB or above
Hard Disk / SSD	256 GB or above
Monitor	15.6-inch display or above
Keyboard	Standard keyboard
Mouse	Standard mouse
Network	Wi-Fi or Ethernet connection
Server System	Computer/Laptop capable of running Node.js and MySQL

The system can be implemented on a standard computer or laptop. A stable network connection is required for communication between the frontend, backend, and database during development and deployment.

4.2 Software Requirements

Operating System: Windows 10/11 for development and testing.

Frontend: React.js with Vite for developing the user interface.

Backend: Node.js and Express.js for server-side development and API handling.

Database: MySQL for storing customer, vehicle, service, and administrator data.

IDE: Visual Studio Code for coding and project development.

Browser: Google Chrome or Microsoft Edge for accessing and testing the application.

Navigation: React Router for navigation between application pages.

API Communication: Axios for communication between frontend and backend.

Authentication: Protected routes for secure administrator access.

Responsive Interface: CSS for creating a user-friendly and responsive interface.

4.3 Technologies Used

Technology	Purpose
React.js	Frontend development and user interface
CSS3	Styling, layout, animations, and responsive design
JavaScript	Frontend functionality and API handling
Node.js	Backend/server-side development
Express.js	Creating REST APIs and handling server requests
MySQL	Database for storing customers, vehicles, service jobs, and admin data
Axios / Fetch API	Connecting React frontend with the backend
React Router	Navigation between different pages
VS Code	Development environment
Git & GitHub	Version control and project management

4.4 Module Description

The **AutoCare Hub** system consists of several modules designed to manage the complete garage service process efficiently. The **Admin Login Module** provides secure access to the system. The **Customer Management Module** allows the admin to add, update, delete, and search customer details. The **Vehicle Management Module** maintains information about customers' vehicles, such as vehicle number, company, and model. The **Service Job Management Module** helps the admin create service jobs by recording service type, vehicle details, service date, and job status. The **Job Status Tracking Module** allows the admin to monitor the progress of each service job, such as pending, in progress, or completed. The **Dashboard Module** provides a quick overview of customers, vehicles, service jobs, and pending services. Finally, the **Profile Module** allows the admin to view and manage profile information. These modules work together to provide a simple and organized system for managing the day-to-day operations of **Auto Motors**.

4.5 User Interface Screens

The **AutoCare Hub** system provides a simple, modern, and user-friendly interface for managing garage operations. The **Login Screen** allows the admin to securely log into the system using valid credentials. The **Dashboard Screen** displays an overview of the system, including the total number of customers, vehicles, service jobs, and pending services. The **Customer Management Screen** allows the admin to add, edit, delete, and search customer records. The **Vehicle Management Screen** is used to manage vehicle details such as vehicle number, company, model, and owner. The **Service Jobs Screen** allows the admin to create and manage service jobs and track their current status. The **Profile Screen** displays the admin's profile information and provides options to update account details. A **Navigation Sidebar and Navbar** are provided throughout the system to make it easy to move between different screens.

4.6 Sample Source Code

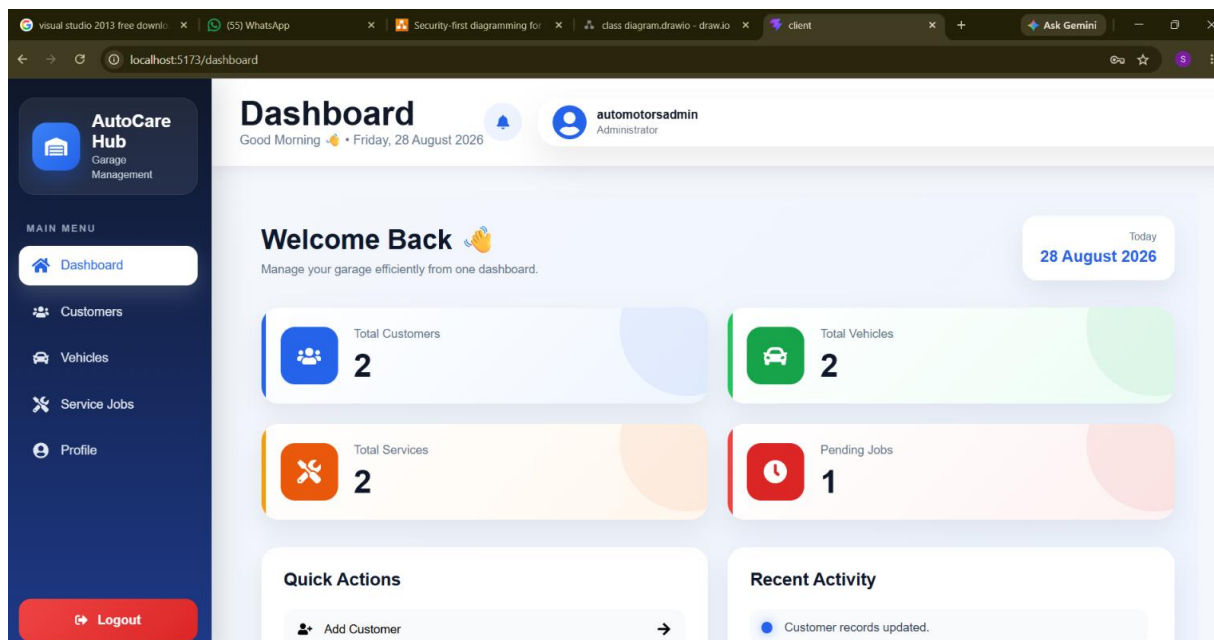
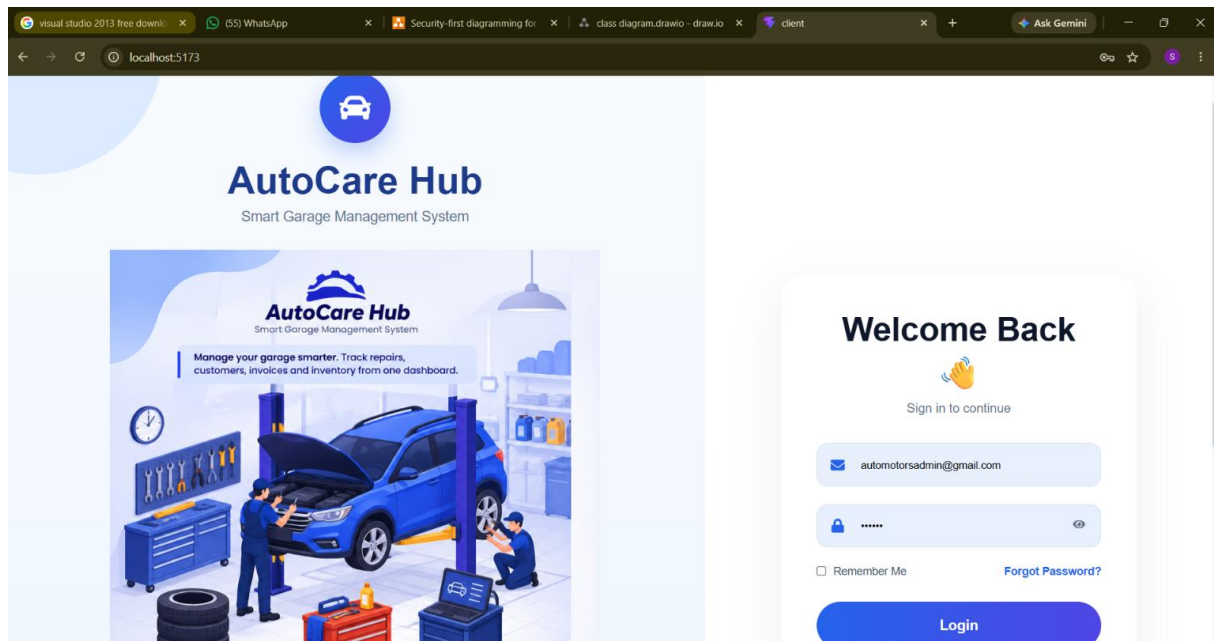
```
App.jsx client http://localhost:5176/dashboard client http://localhost:5175/ # Layout.css client http://localhost:5175/
client > src > App.jsx > default
1  import { Routes, Route } from "react-router-dom";
2
3  import Login from "../pages/Login";
4  import Dashboard from "../pages/Dashboard";
5  import Customers from "../pages/Customers";
6  import Vehicles from "../pages/Vehicles";
7  import ServiceJobs from "../pages/ServiceJobs";
8  import Profile from "../pages/Profile";
9
10 import ProtectedRoute from "../components/ProtectedRoute";
11
12 function App() {
13   return (
14     <Routes>
15       <Route path="/" element={<Login />} />
16
17       <Route
18         path="/dashboard"
19         element={
20           <ProtectedRoute>
21             <Dashboard />
22           </ProtectedRoute>
23         }
24       />
25
26       <Route
27         path="/customers"
28         element={
29           <ProtectedRoute>
30             <Customers />
31           </ProtectedRoute>
32         }
33       />
34     </Routes>
35   );
36 }
```

CHAPTER 5: TESTING AND RESULTS

5.1 Test Cases

Test cases are used to check whether the **AutoCare Hub** system functions correctly according to the specified requirements. Different test cases are performed for modules such as **Admin Login, Customer Management, Vehicle Management, Service Job Management, Dashboard, and Profile**. The system is tested for valid and invalid login credentials, adding, updating, deleting, and searching customer and vehicle records, creating service jobs, updating service status, and retrieving data from the MySQL database. API requests and database connectivity are also tested to ensure proper communication between the frontend and backend. These test cases help identify errors, improve system reliability, and ensure that the application provides accurate and expected results.

5.2 Results and Outputs



AutoCare Hub
Garage Management

MAIN MENU

Dashboard

Customers

Vehicles

Service Jobs

Profile

Logout

Customers

Good Morning 🌞 • Friday, 28 August 2026

automotorsadmin
Administrator

Add Customer

2
Total Customers

Search customer...

Customer	Phone	Address	Actions
A Afsha ID #12	9136378729	Malad	
A Atif ID #13	8657907469	Malad	

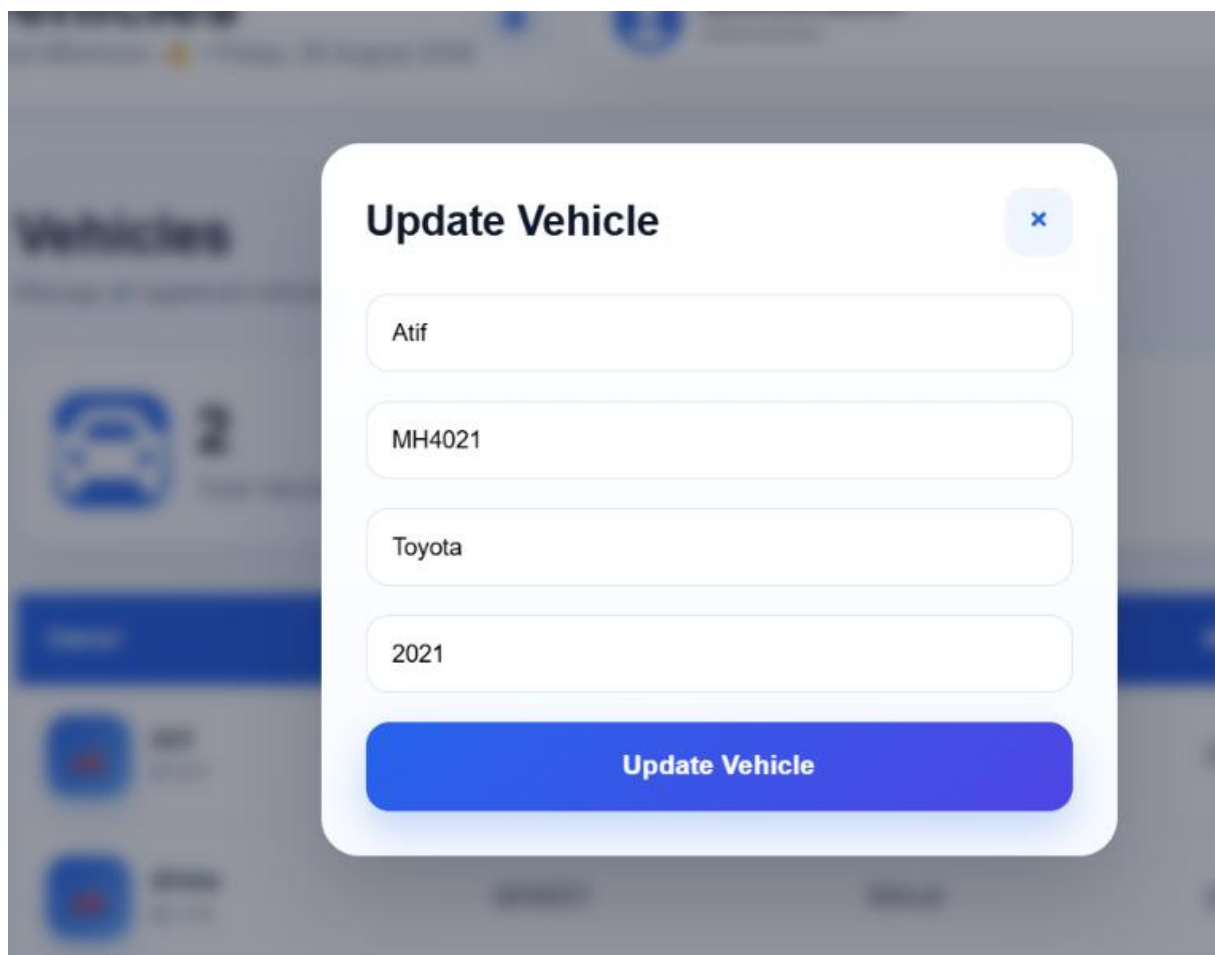
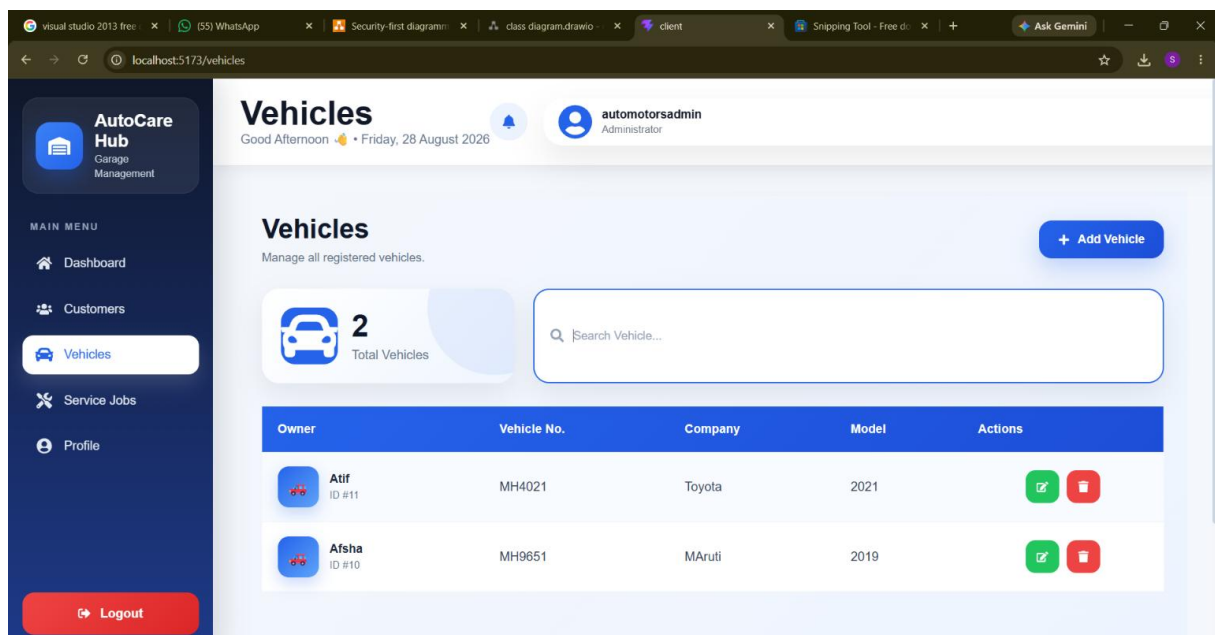
Add Customer

Customer Name

Phone Number

Address

Save Customer



**AutoCare Hub**
Garage Management

MAIN MENU

- Dashboard
- Customers
- Vehicles
- Service Jobs**
- Profile

Logout

Service Jobs

Good Afternoon 🌞 • Friday, 28 August 2026

automotorsadmin
Administrator

+ Create Service

2
Total Jobs

1
Completed

0
In Progress

1
Pending

Search Service...

Vehicle	Service Type	Date	Status	Actions
 MH9651 Vehicle ID #10	Servicing	2026-08-23	Pending	 
 MH4021 Vehicle ID #11	mechanic	2026-08-27	Completed	 

Create Service

Select Vehicle

Service Type

dd-mm-yyyy

Pending

Save Service

Update Service

MH9651

Select Vehicle

MH4021

MH9651

23-08-2026

Pending

Update Service

Update Service

MH9651

Servicing

23-08-2026

August, 2026

↑

↓

Su	Mo	Tu	We	Th	Fr	Sa
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31	1	2	3	4	5

Clear

Today

Update Service

Add Customer

Customer Name

Phone Number

Address

Save Customer

Add Vehicle

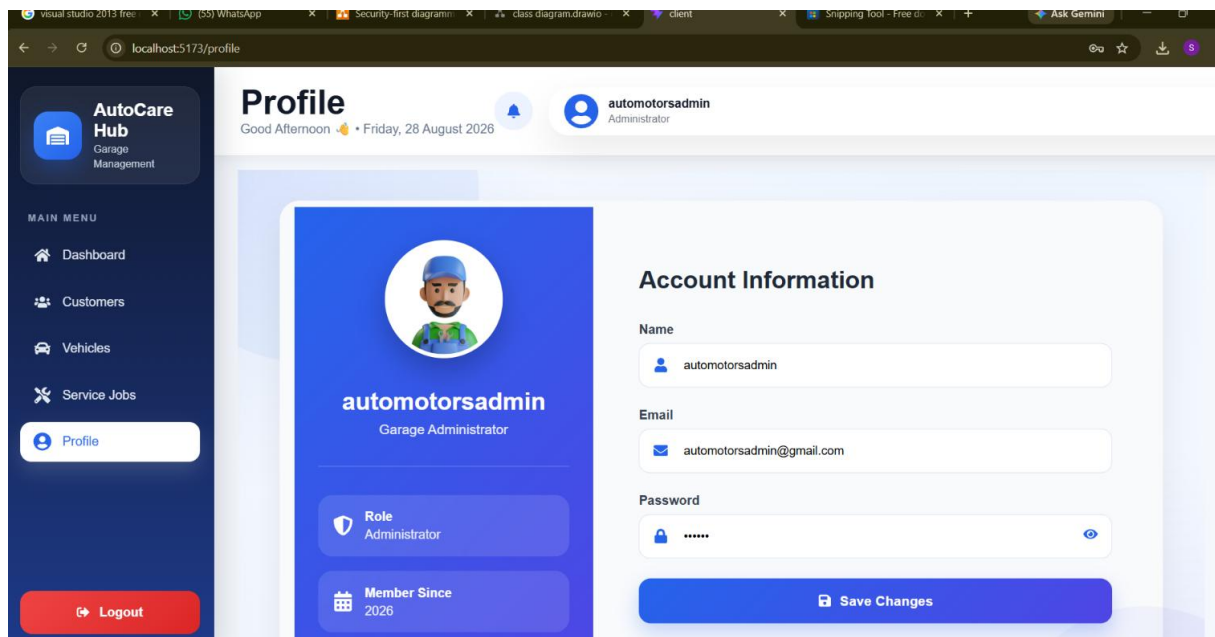
Owner Name

Vehicle Number

Company

Model

Save Vehicle



CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The **AutoCare Hub** project is a web-based garage management system developed to simplify and improve the management of four-wheeler service operations. The system provides a centralized platform where the admin can manage important information related to customers, vehicles, and service jobs. It helps in maintaining accurate customer and vehicle records, creating new service jobs, updating service details, and tracking the status of vehicles throughout the servicing process.

The system is developed using modern technologies such as **React.js, JavaScript, CSS, Node.js, Express.js, and MySQL**. The React-based user interface provides a simple and interactive experience, while the Node.js and Express.js backend handles requests and communicates with the MySQL database. This integration allows information to be stored, retrieved, updated, and deleted efficiently.

AutoCare Hub reduces the dependency on manual paperwork and helps minimize errors in maintaining service records. The dashboard also provides a quick overview of important information, making it easier for the admin to monitor ongoing and pending services. The system improves the overall organization, efficiency, and accessibility of garage operations.

Overall, **AutoCare Hub** provides a reliable, organized, and user-friendly solution for managing the day-to-day activities of **Auto Motors**. The project successfully demonstrates how a full-stack web application can be used to solve real-world problems in the automobile servicing industry. In the future, the system can be enhanced by adding features such as **customer login, online service booking, service history, invoice generation, payment integration, notifications, and reports**, making it more useful for both garage administrators and customers.

6.2 Learning Outcomes

Learning:

During the development of the **AutoCare Hub** project, various technical and practical skills were gained. The project provided hands-on experience in developing a **full-stack web application** using React.js, JavaScript, CSS, Node.js, Express.js, and MySQL. It helped in understanding how frontend and backend components communicate through APIs and how data is stored and managed in a relational database. The project also improved knowledge of **CRUD operations, database connectivity, REST APIs, form handling, authentication, routing, and responsive user interface design**. Along with technical skills, the project provided experience in identifying real-world problems, planning system modules, testing the application, debugging errors, and managing a complete software project.

Outcomes:

The main outcome of the **AutoCare Hub** project is a functional and user-friendly garage management system that helps organize and manage four-wheeler service operations. The system allows the admin to maintain **customer and vehicle records, create and manage service jobs, track service status, and view important information through the dashboard**. It reduces manual paperwork, minimizes data management errors, and makes information easier to access and update. The project also demonstrates the practical application of full-stack development concepts and provides a foundation for future enhancements such as **online service booking, customer login, service history, invoice generation, payment integration, notifications, and detailed reports**.

6.3 Future Scope

The **AutoCare Hub** project can be further enhanced with additional features to make the system more useful, secure, and convenient for both garage administrators and customers. In the future, a **customer login and registration module** can be added so that customers can directly access their vehicle and service information. An **online service booking system** can allow customers to select a service and book an appointment based on available time slots. The system can also include **service history**, allowing customers and administrators to view previous services performed on a vehicle.

Other improvements can include **automatic invoice generation and online payment integration** for easier billing and payment management. **Email or SMS notifications** can be added to inform customers about booking confirmations, service progress, and vehicle completion. The dashboard can be enhanced with **reports, charts, and analytics** to help the garage monitor its performance and service activities. Additional features such as **spare-parts inventory management, employee/mechanic management, customer feedback and ratings, and automated service reminders** can also be introduced.

In the future, the system could be deployed on a cloud platform so that it can be accessed securely from different locations and devices. These enhancements would help transform AutoCare Hub from a basic garage management system into a **complete digital automobile service management platform**.

CHAPTER 7: REFERENCES

- **React.js Documentation** – Official documentation for React.js, used for developing the frontend and user interface.

[React Documentation](#)

- **Node.js Documentation** – Official documentation used for understanding server-side JavaScript and backend development.

[Node.js Documentation](#)

- **Express.js Documentation** – Reference for developing REST APIs and handling HTTP requests in the backend.

[Express.js Documentation](#)

- **MySQL Documentation** – Reference for database creation, SQL queries, database management, and connectivity.

[MySQL Documentation](#)

- **MDN Web Docs** – Used as a reference for HTML, CSS, JavaScript, HTTP, and web development concepts.

[MDN Web Docs](#)

- **W3Schools** – Used for learning and reference related to HTML, CSS, JavaScript, SQL, and web development concepts.

[W3Schools](#)

- **GitHub Documentation** – Reference for version control, repository management, and project development using Git and GitHub.

[GitHub Docs](#)